

PhytoGraph — cycles 17-19

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PhytoGraph — cycles 17-19

Abstract

Cycles 17-19 converted the fork `cc044bf40be3` Track 2, Track 3, and Track 5 Wave 4 outputs from branch-local results into root-workspace integration artifacts. The integration did not start new biological research, refit instruments, fetch new sources, run audit-level validation, or promote any prediction into the master ledgers.

The main result is procedural closure for the fork merge: Track 2, Track 3, and Track 5 validation/ablation packages are now represented in the root workspace, cross-branch consistency tests exist, and the Track 3 milestone namespace drift was repaired by adding the explicit `M4.A-track3-convergence-confounds child` milestone to `plan_of_record.md`. The master `prediction_ledger.tsv` and `speculation_ledger.tsv` remain header-only.

The supplied record contains one direct worker session for cycle 17 (`511704ab-adb6-4c75-b42a-1ba50e78800a`). No separate cycle 18 or cycle 19 researcher, worker, or auditor sessions were supplied or found in the session catalog search for this report. This report therefore treats cycles 18-19 as gaps in the supplied record, not as additional completed work.

Introduction

The previous reporting window closed with Barrier 3 validated and the fork `cc044bf40be3` producing Track 2, Track 3, and Track 5 Wave 4 validation/ablation outputs. In PhytoGraph, a “barrier” is a synchronization point where downstream work is allowed only after schema, provenance, validation, and ledger invariants are checked. Barrier 4 remains the future step that can reconcile track-local validation outputs into the master prediction and speculation ledgers.

Cycles 17-19 focused on integration rather than discovery. The worker session `511704ab-adb6-4c75-b42a-1ba50e78800a`, created on 2026-05-18, reports that fork `cc044bf40be3` was integrated into the main workspace. The generated integration artifact is `reports/wave4_postmerge_integration.md`, and the new test artifact is `tests/test_wave4_postmerge_integration.py`.

A numbering mismatch is present in the artifacts: the integration report frontmatter labels the work as cycle 14, while the reporter input defines this reporting range as cycles 17-19. This report follows the supplied `cycle_range` while preserving artifact metadata as implementation detail.

Approach

The integration cycle read the fork outputs and carried their decisions into the root workspace without changing their scientific status. The worker session and audit-report input list the same core actions:

- Added `reports/wave4_postmerge_integration.md`.
- Added `tests/test_wave4_postmerge_integration.py`.
- Updated `plan_of_record.md` to include the explicit Track 3 child milestone M4.A-track3-convergence-confounds.
- Appended two root ledger events to `promise_ledger.jsonl`: `c1d6b9bb-01ef-44df-8e10-c3c87f31738b` and `0c3f897b-eb7e-4071-9b55-0871da7a73fd`.

The integration test checks five boundaries: Track 2 outcome counts, Track 3 milestone registration, Track 3 non-promotion to the master ledger, Track 5 temporal/source-ablation counts, and header-only master ledgers.

Findings

Root Integration Completed

The root integration report states that fork `cc044bf40be3` Track 2, Track 3, and Track 5 outputs were integrated as track-local Wave 4 validation/ablation packages. These packages remain local evidence for their tracks. They are not master-ledger prediction promotions.

The carried-forward decisions are:

Track	Integrated artifact	Carried-forward decision
Track 2	<code>tracks/track2/reports/track2_wave4_validation_closure.md</code>	H2 is not supported under current accepted-key, source, and living-megafauna controls: 0 validated held-outs, 1 falsified under ablation, 6 data-limited, and 1 insufficient-support.
Track 3	<code>tracks/track3/reports/track3_wave4_validation_ablation.md</code>	H3 remains data-limited. drupe and capsule are pending convergence-prior rows only and do not enter the master ledger.
Track 5	<code>tracks/track5/reports/track5_wave4_temporal_source_closure.md</code>	H5 is not validated under frozen inputs. The Duke/source ablation is carried forward as a validated source-bias null result.

Track 3 Namespace Drift Was Repaired

The integration found one mismatch: the Track 3 branch had recorded its convergence/confound work under `_plan/wave4-track3-validation-ablation-branch` because the exact milestone `M4.A-track3-convergence-confounds` was not yet explicit in `plan_of_record.md`.

Cycle 17 repaired this by adding `M4.A-track3-convergence-confounds` to the Wave 4 milestone table. The root ledger event `0c3f897b-eb7e-4071-9b55-0871da7a73fd` supersedes the branch-local Track 3 event `b1d1cfa7-9289-4ce5-a7c8-cc044bf40202` for namespace purposes. The scientific status did not change: H3 remains data-limited, and Barrier 4 reconciliation remains pending.

Validation And Barrier Checks Passed

The worker session reports the following checks:

Check	Result
Track 2/3/5 branch tests plus post-merge integration test	20 passed in 50.27s
Barrier 3 Atlas instrument contract	PASS: 60,000 pages, 6 tracks
Barrier 2 track enrichment conformance	PASS: 6 tracks checked
Barrier 1 substrate validation	PASS: 363,237 nodes, 641,183 retained hyperedges
promise_check	exit 0, 163 events
org_check	exit 0 with inherited root-layout warnings
prediction_ledger.tsv	1 line, header only
speculation_ledger.tsv	1 line, header only

The worker also noted that `validate_barrier2_track_enrichment.py` rewrote `data/barrier2_track_enrichment_conformance.json`, which was expected validator behavior.

Master Ledgers Remained Closed

The integration preserved the central evidence boundary: no Track 2, Track 3, or Track 5 row was promoted to the master prediction or speculation ledger. Both files remain header-only.

This matters because PhytoGraph distinguishes track-local evidence from cross-track campaign predictions. A track-local row can be useful and tested, but it does not become a campaign-level prediction until Barrier 4 reconciliation assigns it a status in the master ledger.

Discussion

Cycles 17-19 did not change the biological interpretation of the Track 2, Track 3, or Track 5 Wave 4 results. Instead, they made those branch outputs visible and testable in the root workspace.

The integration stabilizes three outcomes for downstream work:

- Track 2's H2 recovery target is not supported under the current controls. The integration carries forward the null/data-limited result rather than weakening it into a speculative claim.
- Track 3's H3 convergence statistic remains data-limited. The drupe and capsule rows are local pending priors, not validated convergence claims.

- Track 5's H5 temporal validation is not supported under frozen inputs, while the Duke/source ablation remains a validated source-bias null result.

The integration also narrows the next procedural step. Barrier 4 is still required before any master-ledger reconciliation. The current root state has tests and ledger events proving integration consistency, but it does not yet have audit-level validation of the post-merge package.

Open Questions

1. No audit-level validation session was supplied for cycles 17-19. The integration was tested and validator-clean, but the input itself marks audit-level validation as deferred.
2. Barrier 4 reconciliation remains pending. The master `prediction_ledger.tsv` and `speculation_ledger.tsv` are still header-only, so no campaign-level prediction status has been assigned to the Track 2/3/5 Wave 4 outputs.
3. No separate cycle 18 or cycle 19 sessions were supplied or found during the session search. If work occurred in those cycles outside the supplied session catalog, it is not represented in this report.
4. Inherited `promise_check` and `org_check` warnings remain. The worker characterized them as legacy warnings about historical noncanonical paths, missing legacy report/manager artifacts, root-layout warnings, and unrelated plan milestones without direct events.

References

No new external references were cited by the cycle 17-19 integration work. The integrated Track 2, Track 3, and Track 5 branch artifacts rely on references already accumulated in `REFERENCES.md`, but this report cites only the integration record and local artifacts for the reported cycle range.

Appendix: Implementation Details

Source Inventory

Source ID	Date	Contents	Role in timeline
511704ab-adb6-4c75-b42a-1ba50e78800a	2026-05-18T05:44:18Z	Worker summary of post-merge integration, files added/updated, checks run, results, and deferred work.	Direct cycle 17 source session.
audit_report input	supplied with task	Same post-merge integration summary as the worker session, including tests, validators, interpretation, and sufficiency check.	Reporter-supplied audit trail for the cycle range.
reports/wave4_postmerge_integration.md	2026-05-18T11:00:00Z in artifact frontmatter	Root integration report for fork cc044bf40be3.	Primary integration artifact.

Source ID	Date	Contents	Role in timeline
tests/test_wave4_postmerge_integration.py	2026-05-18T11:00:00Z in artifact header	Focused tests for Track 2/3/5 integrated outcomes and master-ledger invariants.	Verifies integration consistency.
plan_of_record.md	current workspace artifact	Includes explicit M4.A-track3-convergence-confounds row and explanatory namespace repair note.	Records Track 3 milestone repair.
promise_ledger.jsonl	163 lines after integration	Root events c1d6b9bb-01ef-44df-8e10-c3c87f31738b and 0c3f897b-eb7e-4071-9b55-0871da7a73fd.	Registers integration and Track 3 milestone repair.
MANIFEST.md	updated for this report	Current file inventory, counts, cross-references, and preserved ## Key Files section.	Quick-reference snapshot for future workers.

Code Organization

File	Lines	Purpose
tests/test_wave4_postmerge_integration.py	74	Checks integrated Track 2/3/5 outcome counts, Track 3 milestone registration, and header-only master ledgers.
tracks/track2/scripts/track2_wave4_validation_closure.py	321	Generates Track 2 held-out validation closure outcomes.
tracks/track3/scripts/validate_wave4_convergence.py	378	Generates Track 3 Wave 4 convergence/confound outcomes.
tracks/track5/scripts/build_wave4_temporal_source_closure.py	208	Generates Track 5 temporal/source closure outcomes.

Data And Report Artifacts

Artifact	Size	Role
reports/wave4_postmerge_integration.md	31 lines	Root integration report.

Artifact	Size	Role
tracks/track2/data/track2_wave4_validation_outcomes.tsv	8 data rows	Track 2 held-out outcomes.
tracks/track3/data/track3_wave4_validation_outcomes.tsv	16 data rows	Track 3 trait outcomes.
tracks/track3/data/track3_wave4_validation_summary.json	78 lines	Track 3 H3 and confound summary.
tracks/track5/data/track5_wave4_validation_outcomes.tsv	13 data rows	Track 5 temporal and source-ablation outcomes.
prediction_ledger.tsv	0 data rows	Master prediction ledger remains header-only.
speculation_ledger.tsv	0 data rows	Master speculation ledger remains header-only.

Test Results

The worker reported:

```
20 passed in 50.27s
Barrier 3 PASS: 60000 pages, 6 tracks
Barrier 2 PASS: 6 tracks checked
Barrier 1 PASS: 363237 nodes, 641183 retained hyperedges
promise_check exit 0: 163 events
org_check exit 0 with inherited root-layout warnings
```

Cross-Reference Map

Origin	Consuming artifact	Flow
Fork cc044bf40be3 Track 2 output	reports/wave4_postmerge_integration.md, tests/test_wave4_postmerge_integration.py	Track 2 H2 outcome counts are carried into root integration.
Fork cc044bf40be3 Track 3 output	plan_of_record.md, promise_ledger.jsonl	Track 3 surrogate branch milestone is reconciled to M4.A-track3-convergence-confounds.
Fork cc044bf40be3 Track 5 output	reports/wave4_postmerge_integration.md, tests/test_wave4_postmerge_integration.py	Track 5 H5 and Duke/source-ablation outcomes are carried into root integration.
prediction_ledger.tsv, speculation_ledger.tsv	Future Barrier 4 reconciliation	Both remain header-only until a controlled master-ledger reconciliation step.

Manifest Update

MANIFEST.md was updated for report_cycles_17-19. Its existing ## Key Files section was preserved verbatim, and the new snapshot records the post-merge integration files, integrated Track 2/3/5 packages, validation counts, ledger status, and cross-reference map.